SHAGGAR INSTITUTE OF TECHNOLOGY
Excellence in Science and Technology

COMP 1210 - Course Syllabus

1. Basic Course Information

Course Code and Title: COMP 1210 - Introduction to Programming

Credit Hours: 3

Instructor(s): Dr. Yonatan Ashenafi, Andrea Wu

Support Instructors: Yoanna Turura and Selam Brook

Semester/Quarter: Winter 2026

Prerequisites: N/A

Co-requisites: N/A

Class Schedule: Monday/Wednesday (3 - 4:30 pm in the afternoon EAT)

Breakout Classes: Tuesday and Thursday (6:00-7:00 pm at night EAT), Wednesday and Friday (8:00 - 9:00 pm in the afternoon EAT)

Office Hours: To be determined, feel free to email instructors

Contact Information: yonatanashenafi0@gmail.com, andreaowu@gmail.com

2. Course Description

This course provides a hands-on introduction to programming using Python, designed around real-world, project-based learning. Students will develop foundational computational and algorithmic thinking skills. They will also build meaningful applications that will solve problems in STEM. Topics include data types, control structures (conditionals and loops), modularity (functions and classes), basic data structures (lists, dictionaries, tuples), file input/output, and an introduction to modeling physical and biological systems, analyzing data, visualizing results, and implementing computational solutions aligned with engineering best practices. Emphasis is placed on readability and efficiency. No prior programming experience is required.

3. Course Objectives

Upon successful completion of the course, students will be able to:

1. Understand the role computation can play in solving problems.

2. Demonstrate proficiency in Python programming fundamentals.

3. Apply computation thinking to model scientific and engineering problems.

4. Use Python libraries (e.g., NumPy, Pandas, Matplotlib, SciPy) for numerical computing and data analysis.

5. Implement reproducible workflows using Jupyter notebooks and version control.

6. Communicate computational findings clearly through visualizations and technical documentation.

4. Student Learning Outcomes (ABET Outcomes 1–7)

Upon successful completion of the course, students will be able to:

1. Identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

2. Apply engineering design to produce solutions that meet specified needs with consideration for public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.

3. Communicate effectively with a range of audiences.

4. Recognize ethical and professional responsibilities in engineering situations and make informed judgments.

5. Function effectively on a team to provide leadership, create an inclusive environment, establish goals, plan tasks, and meet objectives.

6. Develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.

7. Acquire and apply new knowledge as needed, using appropriate learning strategies.

5. Mapping to Program Outcomes

ABET Outcomes	Course Objectives
1. Identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.	2, 3 ,4
2. To produce solutions that meet specified needs with consideration for public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.	2, 4, 5
3. Communicate effectively with a range of audiences.	6

4. Recognize ethical and professional responsibilities in engineering situations and make informed judgments.	5, 6
5. Function effectively on a team to provide leadership, create an inclusive environment, establish goals, plan tasks, and meet objectives.	4, 6
6. Develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.	3, 4
7. Acquire and apply new knowledge as needed, using appropriate learning strategies.	1, 3, 5

6. Textbooks and References

Textbooks: Introduction to Computation and Programming Using Python. John V. Guttag. Second Edition, Head-first series

MIT OpenCourseWare: Introduction to Computer Science and Programming in Python

7. Course Topics and Schedule

Topic	Reading	Assignment (s)
	Module 1	
Introduction to computational thinking and programming: data types, arithmetic expressions, operator precedence, variables, input/output, Google Colab, Github Classroom	Guttag, Ch.1; Ch. 2.1, 2.3	Write Python scripts using variables, arithmetic, and variables
	Module 2	
Control Flow and Iteration: comparison / logical operators, conditionals, while loops, strings, for loops	Guttag, Ch. 2.2-2.4; Ch 3.2, 3.3	Implement iterative approximation algorithm (e.g., root finding)
	Module 3	
Functions and Modular Design: function def, parameters and return values, modules and imports, scope, pass by value vs reference, (recursion)	Guttag, Ch. 4.1, 4.3, 4.5	Write functions with given parameters

Module 4		
***Debugging and testing*:** how to debug, why testing, unit / integration tests	Guttag, Ch. 6.1, 6.2	Write tests for functions
Module 5		
Data Structures: lists, mutability, 2D lists, dictionaries, nested dictionaries, sets	Guttag, Ch. 5.2, 5.3, 5.5, 5.6	Build a structured data-processing program using lists and dictionaries
Module 6		
Term Project Initiation: problem formulation, MVP, model design, implementation, validation, Github		Project proposal
Module 7		
Numerical Computing and Visualization: NumPy arrays, vectorization, Matplotlib visualization, interpreting plots	Guttag, Ch. 11	Analyze and visualize a dataset that does X (we can have choices for X to cover diverse disciplines)
Module 8		
Introduction to algorithms: algorithmic thinking, pseudocode, flowcharts, and basic efficiency (time/space) ideas	Guttag, Ch. 9	Design and analyze an algorithm to solve a problem, represent the solution using pseudocode/flowcharts, and trace algorithmic complexity
Module 9		
Ethics in Computational Modeling and Final Project Development: responsible AI and modeling ethics, data integrity, documentation standards		Project progress report
Module 10		
Final Examination and Project Presentations: ● exam (conceptual understanding, code tracing, short design questions,		Project deliverables, presentation, and demo

interpretation of computational results) ● project (code repo, technical report, oral presentation and demo)		

8. Breakout Sessions

In your breakout sessions, you will work on problem sets, homework assignments, and review concepts discussed in lecture. We will be using the following tools:

- Google Colab: for all coding work
- Github Classroom: for homework/project management
- Python 3.14.2
- Jupyter Notebook: for local usage

9. Grading Policy

Assignment/Activities	Total Points	Percentage of Grade
Weekly Homeworks	140	10%
Quizzes	100	30%
Final Exam	100	30%
Final Project	100	30%

Assignments:

- Due date and time: all assignments are due on **Sunday night by 11:59 pm EAT**
- The problems are posted weekly on the course page of the SIT's LMS.
- Submission format: will be specified within the homework instructions.

Quizzes:

- Administered during class in-person and graded.

Final Project:

- The purpose of students' projects is to enhance collaboration and foster a team spirit while exploring new topics outside the classroom. The final project will be done in teams of 4 or 5 students.
- Project proposals will be due on the respective Sunday night by 11:59 PM EAT.
- More details will be provided through a project guideline document.

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10. Academic Integrity

For SIT's policies about Academic Integrity, Class Attendance, Classroom Behavior, Accessibility and Student Support, and Safety Considerations, please read the SIT Student Manual.

The following policies are essential for this course.

Academic Honesty

Cheating on any assignment, quiz, or exam undermines the integrity of the academic system and will not be tolerated. Cheating in this class is defined as: copying code from peers or online resources (including AI / LLMs and resources found through search engines). Changing variable / function / argument names from copied code still counts as copying. Do not ask for answers, do not give answers.

Attendance

Attendance will be taken at every meeting. Students are expected to attend every online class and to be on time. If you are absent from any online and/or in person class for any reason, it is your responsibility to find out what you missed, including any announcements.

Classroom Behavior

You are expected to be in the classroom before the start of class time and leave at the end of the class time. Excessive talks without the permission of the instructors is prohibited. The use of any electronic device is allowed for the lecture and class activities only.

AI Usage Policy

AI tools (e.g., ChatGPT, Copilot, Gemini, Claude) are part of modern computing. This course recognizes their relevance but prioritizes independent learning and genuine skill development. Students may use AI tools to clarify programming concepts, interpret documentation, or understand error messages. Students may NOT use it to write or debug any code. This means students may not submit AI-generated code, copy solutions, use AI during quizzes or exams, or rely on AI for take-home assignments. If AI is used, students must include the following statement in their submission: "AI Usage Statement: I used [tool name] to assist with [brief description], and here is the conversation: [link]. I confirm that I understand and can explain all submitted code." Undisclosed AI use constitutes an academic integrity violation. The instructional team will conduct brief oral code checks on randomly selected students to verify understanding. AI should support your learning, not replace the critical-thinking and problem-solving required to master programming.

11. Accessibility and Student Support

Information on disability services, tutoring, counseling, and academic support resources.

12. Safety Considerations

Outline safety guidelines for labs, fieldwork, or other activities.